

National University of Computer and Emerging Sciences



Lab Manual 11 Object Oriented Programming

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Objectives:

After performing this lab, students shall be able to:

- ✓ Understand Polymorphism
- ✓ Virtual and Pure virtual functions
- ✓ Abstract classes
- ✓ Basic Templates

TASK-1: Templates

Write a simple function template for predicate function `isEqualTo` that compares its two arguments of the same type with the equality operator (`==`) and returns true if they are equal and false otherwise. Use this function template in a program that calls `isEqualTo` only with a variety of fundamental types.

TASK-2: Abstract Classes and Polymorphism

Task 2.1:

- Create an **abstract** class called Account.
- It has data member:
 - Private Account Number.
 - Private Account Balance.
- Add suitable setter/getter for data.
- Add `Debit(float)`, `Credit(float)` as member functions (**Pure Virtual**).
- Add `Print()` function (**Virtual**)
 - Override `Print` `Debit` and `Credit` functions according to derived classes.

Task 2.2:

- Create a class called `CurrentAccount` i-e: `CurrentAccount`(is-a) `Account`
- It has data member:
 - Service Charges (To be charged during credit if account balance is less than min balance)
 - Minimum Balance
- Override `print()` as created in above class which displays:
 - Account Number, Account Balance, Minimum Balance, Service Charges
- Modify the definition of the `print()` so that it displays a suitable message containing above info.
- Similarly override `credit(float)`, `debit(float)` functions such that `credit(float)` simply add amount to the Account Balance and `debit(float)` checks if the amount to be debited is within the range of Account Balance, and further if the amount is account balance is less than min balance standard charges would also be deducted.
- Create a class called `SavingAccount` i-e: `SavingAccount` (is-a) `Account`
- It has data member:
 - Interest Rate.
- Override `print()` as created in parent class which displays:

- Account Number, Account Balance, Interest Rate
- Modify the definition of the print() so that it displays a suitable message containing above info.
- Similarly override credit(float), debit(float) functions such that credit(float) simply add amount to the Account Balance and debit(float) checks if the amount to be debited is within the range of Account Balance.

Taks 2.3:

- In main function:
 - Create an array of Account type Pointers, of size 3. Assign Account object to index 0, CurrentAccount object to index 1 and SavingAccount object to index 2.
 - Have you encountered any problem? Report the problem and change the size and elements of the array accordingly
 - Now Credit and Debit the CurrentAccount and SavingAccount in main function

Task 2.4:

Although things seem to be fine on the surface, there is a problem in the program we just wrote. To observe this problem, we must add destructors for all classes. Paste the following inline definitions of the destructors in their corresponding classes, execute the program and paste the output below.

Your destructors should delete the child classes first before destroying parent class. There should not be any memory leaks.